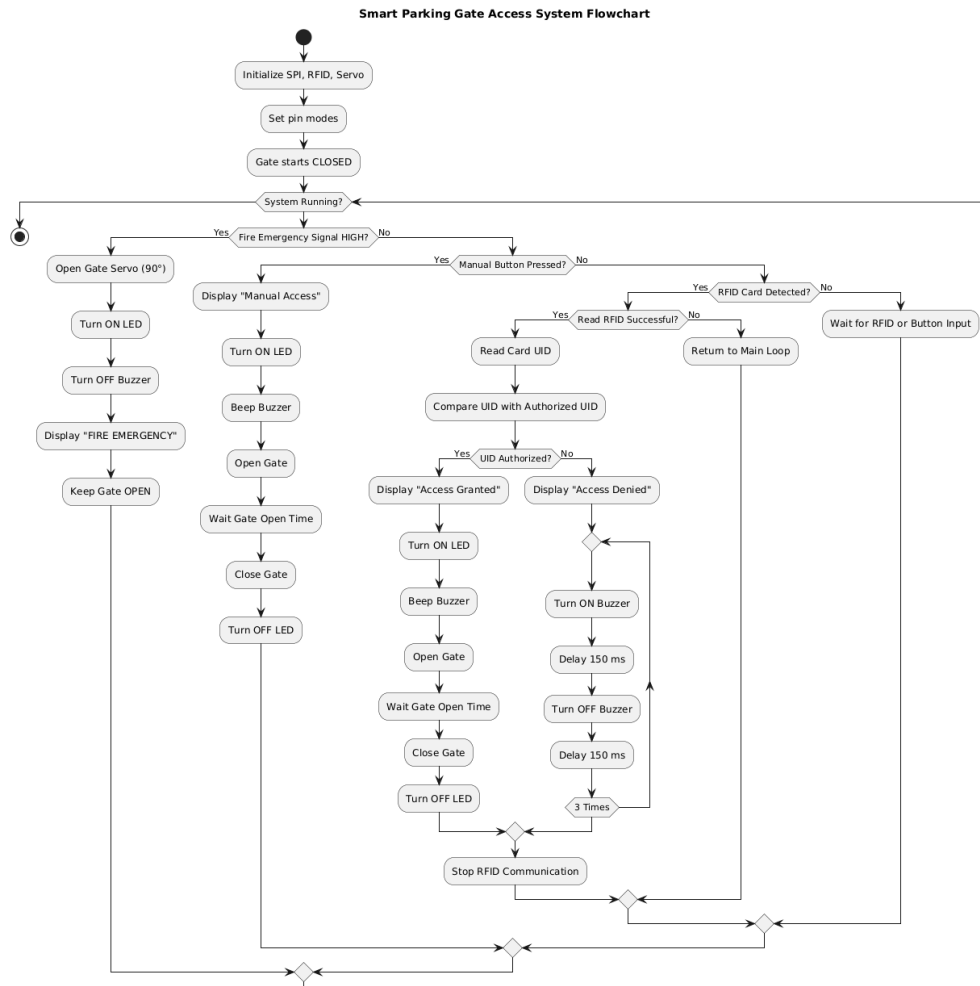


Individual Component Noor Ur Rashid

Input Output Interface Design and Programme Development 1

Smart Home Parking Gate Access System

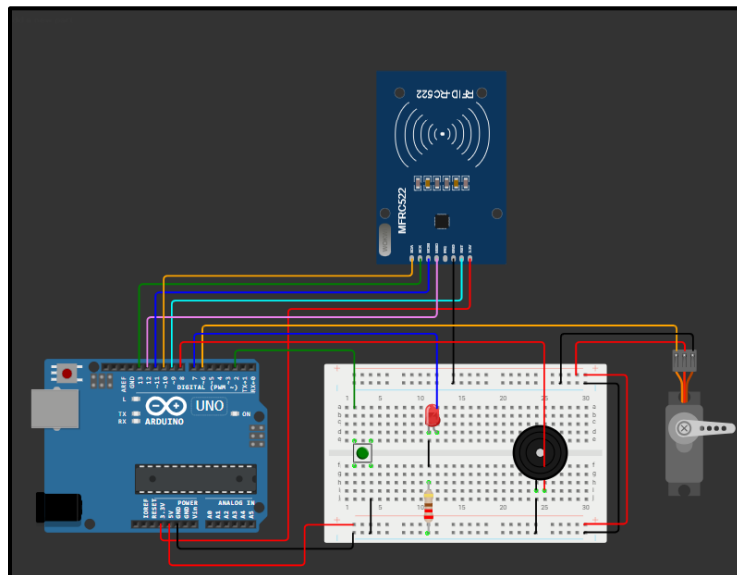


Flowchart of Smart Home Parking Gate Access System

The flowchart starts with the initialization of the servo motor, RFID module, input & output pins. The parking gate is open on power up to ensure access to the site is not denied. Once the system is initialized it is continuously operated with the main loop, monitoring various conditions. The

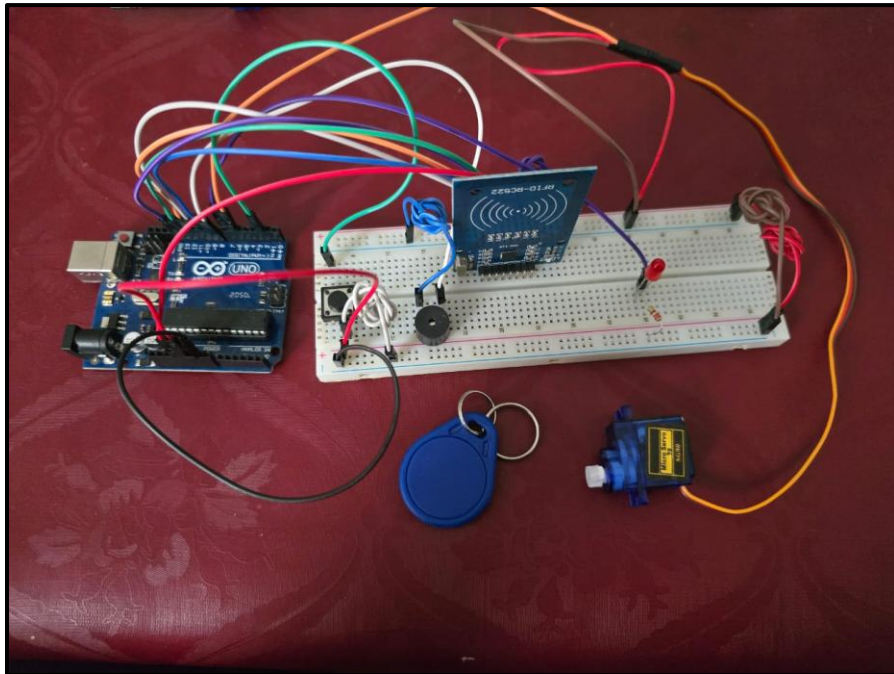
first check is of the fire emergency signal. In case of fire emergency, the normal RFID authentication process is cut, and the gate is opened by rotating the servo motor by 90 degrees. Meanwhile, the LED indicator is lit and the buzzer is off for safe and quick evacuation. Gate remains open until emergency condition has been eliminated. The manual override button is checked if no fire emergency is detected. Press the button and access is provided manually from within the house. The LED turns on, a brief buzzer sounds and then the gate is opened for a set amount of time, followed by a re-opening of the gate automatically after the set time. If the RFID reader doesn't detect a card, the manual button is then checked. If the RFID card is detected, the data on the card is read and the UID stored on the card is compared to the stored authorized UID. When the UID is equal to the authorized RFID card, the gate is opened for a short time, the buzzer sounds a confirmation sound, and then the gate is automatically closed. If the UID is not matched, however, access is denied and a warning buzzer pattern is displayed three times in order to signal an unauthorized access. After every RFID operation, the RFID communication is halted and the system goes back to the main loop and all conditions are repeatedly checked.

Simulation:



Circuit Design Simulation

The simulation diagram is the hardware implementation of Smart Home Parking Gate Access System. Arduino Uno is employed to act as the central controller, and all the components are connected to it. The RFID module reads the UID of the card to open the door, and a push button is added as a manual override that is located inside the house. The gate motion is simulated by connecting a servo motor, which turns to open or close the gate. A visual indicator (LED) is employed and a buzzer is used as an audio feedback to grant access or deny access. A resistor is used to restrict the current in the LED. The Arduino will process input signals, which are sent by the RFID module and the push button, during operation. In case of valid access, the gate is opened and indicators are turned on. Otherwise access is denied and a warning sound generated.



Physical Circuit Setup

Code:

```
1  #include <SPI.h>
2  #include <MFRC522.h>
3  #include <Servo.h>
4
5  #define SS_PIN 10
6  #define RST_PIN 9
7
8  // =====
9  // COMPONENT PINS
10 // =====
11 const int servoPin = 6;
12 const int buttonPin = 2;
13 const int ledPin = 7;
14 const int buzzerPin = 8;
15
16 // 🔥 Fire emergency signal from Eshan
17 const int fireEmergencyPin = A2;
18
19 // =====
20 // RFID SETUP
21 // =====
22 MFRC522 rfid(SS_PIN, RST_PIN);
23 Servo gateServo;
24
25 // =====
26 // AUTHORIZED RFID UID
27 // =====
28 byte authorizedUID[] = {0x73, 0xC2, 0x5F, 0x2D};
29 const byte uidLength = 4;
30
31 // =====
32 // GATE OPEN TIME
33 // =====
34 const unsigned long gateOpenTime = 5000;
35
```

```

36 // =====
37 // SETUP
38 // =====
39 void setup() {
40
41     Serial.begin(9600);
42
43     SPI.begin();
44     rfid.PCD_Init();
45
46     pinMode(buttonPin, INPUT_PULLUP);
47
48     pinMode(ledPin, OUTPUT);
49     pinMode(buzzerPin, OUTPUT);
50
51     pinMode(fireEmergencyPin, INPUT);
52
53     gateServo.attach(servoPin);
54
55     // Gate starts closed
56     gateServo.write(0);
57
58     digitalWrite(ledPin, LOW);
59     digitalWrite(buzzerPin, LOW);
60
61     Serial.println("Smart Home Parking Gate Access System");
62     Serial.println("Scan RFID card or press manual button.");
63 }
64
65 // =====
66 // MAIN LOOP
67 // =====
68 void loop() {
69

```

```

70 // =====
71 // 🔥 FIRE EMERGENCY MODE
72 // =====
73 if (digitalRead(fireEmergencyPin) == HIGH) {
74
75     gateServo.write(90);
76
77     digitalWrite(ledPin, HIGH);
78     digitalWrite(buzzerPin, LOW);
79
80     Serial.println("FIRE EMERGENCY - GATE OPEN");
81
82     return;
83 }
84
85 // =====
86 // MANUAL OVERRIDE BUTTON
87 // =====
88 if (digitalRead(buttonPin) == LOW) {
89
90     Serial.println("Manual button pressed - Gate opening");
91
92     grantAccess();
93
94     delay(300);
95
96     return;
97 }
98
99 // =====
100 // CHECK RFID CARD
101 // =====
102 if (!rfid.PICC_IsNewCardPresent()) {
103     return;
104 }

```

```
105
106 if (!rfid.PICC_ReadCardSerial()) {
107     return;
108 }
109
110 // =====
111 // PRINT CARD UID
112 // =====
113 Serial.print("Card UID:");
114
115 for (byte i = 0; i < rfid.uid.size; i++) {
116
117     Serial.print(" ");
118
119     if (rfid.uid.uidByte[i] < 0x10) {
120         Serial.print("0");
121     }
122
123     Serial.print(rfid.uid.uidByte[i], HEX);
124 }
125
126 Serial.println();
127
128 // =====
129 // CHECK AUTHORIZATION
130 // =====
131 if (isAuthorized(rfid.uid.uidByte, rfid.uid.size)) {
132
133     Serial.println("Access Granted - Gate opening");
134
135     grantAccess();
136
137 } else {
138
139     Serial.println("Access Denied");
```

```

140
141     denyAccess();
142 }
143
144 rfid.PICC_HaltA();
145 rfid.PCD_StopCrypto1();
146 }
147
148 // =====
149 // RFID AUTHORIZATION
150 // =====
151 bool isAuthorized(byte *uid, byte size) {
152
153     if (size != uidLength) {
154         return false;
155     }
156
157     for (byte i = 0; i < uidLength; i++) {
158
159         if (uid[i] != authorizedUID[i]) {
160             return false;
161         }
162     }
163
164     return true;
165 }
166
167 // =====
168 // ACCESS GRANTED
169 // =====
170 void grantAccess() {
171
172     digitalWrite(ledPin, HIGH);

```

```

173
174     // Success beep
175     digitalWrite(buzzerPin, HIGH);
176
177     delay(200);
178
179     digitalWrite(buzzerPin, LOW);
180
181     // Open gate
182     gateServo.write(90);
183
184     Serial.println("Gate is OPEN");
185
186     // Keep gate open
187     delay(gateOpenTime);
188
189     // Close gate
190     gateServo.write(0);
191
192     digitalWrite(ledPin, LOW);
193
194     Serial.println("Gate is CLOSED");
195 }
196
197 // =====
198 // ACCESS DENIED
199 // =====
200 void denyAccess() {
201
202     digitalWrite(ledPin, LOW);
203
204     gateServo.write(0);
205

```

```
206 // Warning beep pattern
207 for (int i = 0; i < 3; i++) {
208     digitalWrite(buzzerPin, HIGH);
209     delay(150);
210     digitalWrite(buzzerPin, LOW);
211     delay(150);
212 }
213 }
214 }
215 }
216 }
217 }
```

Arduino Code for Smart Home Parking Gate Access System

This program was designed to build a Smart Home Parking Gate Access System based on RFID authentication technology, servo motor parking gate system, manual override button and fire alarm integration system. First, the libraries needed for the SPI communication, RFID operation and servo motor control were listed at the top of the code. Communication between the Arduino and the RFID module used the SPI library; the control of the RFID reader is done using the MFRC522 library and the control of the movement of the gate servo motor is done using the Servo library.

Next a number of pins for the various hardware components of the system were identified. The servo motor was connected to the pin #6, the manual override button was connected to the pin #2, the LED indicator was connected to the pin #7 and the buzzer was connected to pin #8. Moreover, A2 was used as the input pin for the fire emergency signal and in case of a fire alarm, a signal from Eshan's fire safety system could be received at this pin.

After the pin definitions, an RFID object and a servo object were created. The byte array of the authorized RFID card UID was also embedded within the program. This UID was used to see if the access should be granted or denied when a card was scanned. A gate opening time of 5 seconds was also specified to allow the gate to stay open for a brief period of time before it would close automatically.

In the setup() function, serial communication was enabled at a 9600 baud rate, enabling the monitoring to be carried out via the Serial Monitor. SPI communication and the RFID reader was initialized to make the RFID module ready to operate. The button pin was set as an

INPUT_PULLUP and the LED and buzzer pins were set as outputs. Fire emergency signal pin is set as input pin. The servo motor was connected to the designated servo pin and was properly set to 0 degrees (closed gate) to ensure that the gate is closed at startup. LED and buzzer were also disabled at the beginning.

Everything the main operation of the system was done in loop() function. The first was to constantly monitor the fire emergency signal. The normal security system was automatically overridden during a fire emergency pin. Immediately the gate was opened by setting the servo motor to 90 degrees, the LED indicator was activated and the buzzer was de-activated. This also printed a fire emergency message to the Serial Monitor. The rest of the RFID authentication process was then skipped due to the return statement at the emergency condition.

If no fire emergency was detected, the manual override button was checked. If the button is pressed, then the grantAccess() function is called. This enabled the gate to be unlocked from inside the house without an RFID card. There was also a brief lag added to prevent bouncing of the buttons.

The RFID system was then tested. The first operation by the RFID reader was to check for a new card. In the absence of any card, the system came back to the start of the loop. If a card was detected successfully, the card serial number was read and printed out in hexadecimal on the Serial Monitor. All the bytes of the RFID UID were printed clearly by using a loop.

Next, the scanned RFID UID was compared to the one stored in the allowed UID using the isAuthorized() function. Within this function, the size of the scanned UID was first verified to see if it matched the length stored in UID. The information in each of the bytes of the scanned UID was then compared with the authorized UID, sequentially. The function returned true if all bytes matched, meaning that the card was authorized. Otherwise, false was returned.

GrantAccess() function was called if access was allowed. In this function, the LED indicator was turned on and a short buzzer beep was produced to indicate successful authentication. The gate servomotor was turned to 90° to open the gate and the gate was open for the set duration. Once the delay period had elapsed, the servo motor was then sent back to 0 degrees to reopen the gate and the LED was switched off.

A denyAccess() function was called if access was denied. Here, the LED was turned off, the gate remained closed and after a warning beep pattern was beeped three times, indicating unauthorized access to the gate.

Finally, to avoid communication problems and be ready for the next card scanning, the RFID communication session was terminated after each RFID operation by calling PICC_HaltA() and PCD_StopCrypto1().

Components List:

Components Name	Quantity
Arduino Uno	1
RFID RC522	1
RFID Card / Key Tag	1
Servo Motor (SG90)	1
Push Button	1
LED	1
Piezo Buzzer	1
220 Ω Resistor	1
Breadboard	1
Jumper Wires	As much required

Enhancement 1

The Smart Home Parking Gate Access System was also improved by incorporating it with another group member's Fire and Gas Safety System. The feature "Fire emergency communication" was added in order to activate the emergency signal without having to perform the normal RFID authentication. The fire condition detection system opened the gate when a fire condition was detected to facilitate rapid evacuation and emergency access. This enhancement brought the level of safety and response to emergency situations in the system to a much higher level. Moreover, a

dual access control system was provided. The manual override push button was added as an extra access mechanism to the RFID based system. This feature allowed the gate to be operated in the event of a failure of the RFID system or if the system was unavailable, from within the house. This has resulted in greater reliability and usefulness of the system. Another improvement was to introduce automatic closing of the gates. After access was granted, and the gate was opened by the servo motor, the gate was kept open for a set amount of time and then automatically closed. The enhancement has been added to make the system more convenient and to minimise the risk of the gate being opened by mistake, enhancing the security of the system. In addition, the system was equipped with audio and visual feedback mechanisms. Different indication sounds were given by a buzzer when access was granted or denied or when in emergency conditions and an LED indicator was used to indicate the operational status of the system. With these additions, the system became more user-friendly and efficient, allowing for better communication between the user and the system, and providing efficient and user-friendly smart parking gate system.

Result 1

A simulation experiment of the Smart Home Parking Gate Access System was performed and it was successfully implemented and tested. During the testing process, the RFID module was able to accurately detect and read the UID of the authorized RFID card. The servo motor turned on and the gate opened successfully when the valid RFID card was scanned. Concurrently, the LED indicator was lit up and a short buzzer sound was generated while successful authentication was made. The gate was opened for the predetermined time then automatically closed as programmed. If a RFID card (not authorized) was scanned, access was denied and the gate closed. A warning buzzer pattern was triggered when an invalid access was made. This was effective in proving that the system was able to differentiate between its legitimate users and illegal ones, hence enhancing the security efficiency of the parking gate system. The manual override push button was successfully tested also. It was noticed that the gate would be able to be opened from the inside of the house even if the RFID system was not used. The system's ability to back up has improved the reliability and usability of the system during situations where the RFID card was unavailable or malfunctioning. Additionally, the built-in fire emergency was exercised by sending a fire alarm message from the Fire and Gas Safety System. In normal condition, the normal RFID identity

authentication process was not executed, but when the emergency condition occurs, the normal identity authentication process was automatically skipped and the gate was automatically opened for emergency evacuation, and the rescue was carried out smoothly. The LED indicator continued to lit up and the gate was left open during the emergency state. The result indicated that the integrated system was able to perform security and emergency safety functions effectively.

Discussion 1

The Smart Home Parking Gate Access System function, performance, reliability and integration capability were analysed. The system was able to carry out the intended function properly during the simulation in the form of controlling access to the gates by using RFID authentication. The integration of the RFID reader, servo motor, LED indicator, buzzer and manual override button was accomplished and the overall system was stable for the entire testing period. Access control was secured with the help of RFID technology. An RFID card with an approved UID was only able to start the opening mechanism for the gate, and all other RFID cards were refused access. This contributed to improving the security level of the parking gate system. Also, a manual override push button was added making the system more reliable in case of a failure in the RFID or in emergency situations, the gate can still be operated from the inside of the house. Convenience and safety also were found to be enhanced by the automatic gate closing feature. The automatic closure of the gate after the set time was automatically triggered, reducing the chance of the gate becoming opened accidentally. The use of audio and visual indicators were also successful, with the buzzer and LED giving immediate response to the access and emergency situation. This helped to make the user and the system communicate more, and made it easier to use. In addition, the project included the connection of the Fire and Gas Safety System, which was an added value for the safety of the project. In the cases of fire emergency, the normal RFID authentication procedure was automatically replaced and the gate was opened automatically for emergency evacuation and rescue. The results of this integration showed that the system could not only be used for security purposes but also as a system to provide support in case of emergency. While conducting the analysis, some limitations were identified. Now, a fixed gate opening delay is used in the system which may not be appropriate for all real world scenarios because the time required for a vehicle to enter a parking area may vary based on the type of vehicle. Furthermore, the RFID card has to

be read by the reader within close range, which could be less practical than other long-range wireless methods of access. The project was also experimented primarily on the basis of simulation and hence in the real implementation the environmental factors and hardware restrictions were not fully investigated.